



EX : 2.8

# De Moivre's Theorem and its Applications

Complex Numbers • Concept & Theory Overview



## De Moivre's Theorem

### Abraham De Moivre (1667–1754)

A mathematician who was among the first to use complex numbers in trigonometry. The formula that bears his name helped move trigonometry from pure geometry into algebraic analysis.

### De Moivre's Theorem

For any complex number  $\cos\theta + i \sin\theta$  and any integer  $n$ :

$$(\cos\theta + i \sin\theta)^n = \cos n\theta + i \sin n\theta$$

*This single identity is the engine behind every technique in this section — simplifying powers of complex numbers, solving polynomial equations, and finding roots of complex numbers.*

## Corollaries of De Moivre's Theorem

**(1)  $(\cos\theta + i \sin\theta)^n = \cos n\theta + i \sin n\theta$**

*The theorem itself, for any integer  $n$  (positive, negative, or zero).*

**(2)  $(\cos\theta + i \sin\theta)^{-n} = \cos n\theta - i \sin n\theta$**

*Applying the theorem with a negative index.*

**(3)  $(\cos\theta - i \sin\theta)^n = \cos n\theta - i \sin n\theta$**

*The conjugate form raised to the power  $n$ .*

**(4)  $\sin\theta + i \cos\theta = i (\cos\theta - i \sin\theta)$**

*A useful rewriting used to bring mixed sin/cos expressions into cis-form before applying the theorem.*

## Finding the $n$ th Roots of a Complex Number

### Definition

For a positive integer  $n$ , a complex number  $\omega$  is called an  $n$ th root of  $z$ , written  $\omega = z^{1/n}$ , if  $\omega^n = z$

### Setting up the equation

Let  $\omega = \rho(\cos\phi + i \sin\phi)$  and  $z = r(\cos\theta + i \sin\theta) = r[\cos(\theta+2k\pi) + i \sin(\theta+2k\pi)]$ ,  $k \in \mathbb{Z}$

Since  $\omega^n = z$ , De Moivre's theorem gives:

$$\rho^n(\cos n\phi + i \sin n\phi) = r[\cos(\theta+2k\pi) + i \sin(\theta+2k\pi)]$$

### Comparing moduli and arguments:

$$\rho^n = r \quad \text{and} \quad n\phi = \theta + 2k\pi \quad \Rightarrow \quad \rho = r^{1/n} \quad \text{and} \quad \phi = (\theta+2k\pi)/n$$

## The nth Roots — Formula and Remarks

For  $z = r(\cos\theta + i \sin\theta)$ , the  $n$  distinct  $n$ th roots are:

$$z^{1/n} = r^{1/n} [ \cos((\theta+2k\pi)/n) + i \sin((\theta+2k\pi)/n) ], \quad k = 0, 1, 2, \dots, n-1$$

Although  $k$  can be any integer, only  $k = 0, 1, \dots, n-1$  give distinct roots — beyond that the roots repeat cyclically.

### Remark 1 — General form of De Moivre's Theorem

If  $x$  is rational,  $\cos(x\theta) + i \sin(x\theta)$  is one of the values of  $(\cos\theta + i \sin\theta)^x$ .

### Remark 2 — Polar form of the unit circle

$$z = e^{i\theta} = \cos\theta + i \sin\theta \implies |z|^2 = \cos^2\theta + \sin^2\theta = 1 \implies x^2 + y^2 = 1$$

So  $z = e^{i\theta}$  traces the unit circle (radius 1, centre at the origin).



## The nth Roots of Unity

### Definition

The  $n$  solutions of  $z^n = 1$  are called the  $n$ th roots of unity.

In polar form:  $z^n = \cos(2k\pi) + i \sin(2k\pi) = e^{i2k\pi}$ ,  $k = 0, 1, 2, \dots$

Applying the  $n$ th-root formula gives:

$$z = \cos(2k\pi/n) + i \sin(2k\pi/n) = e^{i2k\pi/n}, \quad k = 0, 1, 2, \dots, n-1$$

Denoting  $\omega = e^{i2\pi/n} = \cos(2\pi/n) + i \sin(2\pi/n)$ , the  $n$  roots of unity are exactly

$$1, \omega, \omega^2, \omega^3, \dots, \omega^{n-1}$$

(since  $\omega^k = e^{i2k\pi/n}$  for  $k = 0, 1, \dots, n-1$ ).

## Geometric Interpretation of the $n$ th Roots of Unity

Because  $|\omega| = 1$ , every power  $\omega^k$  also has modulus 1.

- All  $n$  roots  $1, \omega, \omega^2, \dots, \omega^{n-1}$  lie on the unit circle  $|z| = 1$ , centred at the origin.
- Successive roots have arguments differing by exactly  $2\pi/n$  — so the roots are equally spaced around the circle.
- Plotted in the complex plane, the  $n$  roots form the vertices of a regular polygon of  $n$  sides inscribed in the unit circle.

*Example: the cube roots of unity ( $n=3$ ) sit at  $0^\circ, 120^\circ, 240^\circ$  on the unit circle, forming an equilateral triangle; the 4th roots ( $n=4$ ) sit at  $0^\circ, 90^\circ, 180^\circ, 270^\circ$ , forming a square.*

This geometric interpretation extends to the general  $n$ th roots of any complex number  $z = r(\cos\theta + i\sin\theta)$ : they lie on a circle of radius  $r^{1/n}$ , equally spaced by  $2\pi/n$ .



## Key Properties of the nth Roots of Unity

### (1) Geometric progression

The  $n$  roots  $1, \omega, \omega^2, \dots, \omega^{n-1}$  form a G.P. with common ratio  $\omega$ .

### (2) Sum of the roots is always zero

$1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$  (sum of a G.P. with ratio  $\omega \neq 1$ , using  $\omega^n = 1$ )

### (3) Product of the roots

$$1 \cdot \omega \cdot \omega^2 \cdots \omega^{n-1} = (-1)^{n-1}$$

### (4) Regular polygon on the unit circle

All  $n$  roots lie on  $|z| = 1$ , dividing the circle into  $n$  equal arcs and forming a regular  $n$ -sided polygon.

## Rotation Property and the Symbol $\omega$

### Rotation by multiplication

For any complex number  $z$ , the product  $z \cdot e^{i\theta}$  represents  $z$  rotated by  $\theta$  radians in the counter-clockwise direction about the origin.

This is the tool used to construct points of a regular polygon (e.g. vertices of an equilateral triangle inscribed in a circle) directly from one known vertex.

### Notation: the value of $\omega$ depends on $n$

$$n = 2 : \omega = e^{i\pi} \qquad n = 3 : \omega = e^{i2\pi/3}$$

$$n = 4 : \omega = e^{i\pi/2} \qquad n = 5 : \omega = e^{i2\pi/5}$$

$$n = k : \omega = e^{i2\pi/k} \quad \text{— the general } n\text{th root of unity generator}$$



# Summary

- De Moivre's Theorem:  $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ , with its 4 corollaries
- $n$ th roots of a complex number:  $z^{1/n} = r^{1/n}[\cos((\theta+2k\pi)/n) + i\sin((\theta+2k\pi)/n)]$
- $n$ th roots of unity:  $z = e^{i2k\pi/n} = 1, \omega, \omega^2, \dots, \omega^{n-1}$
- Sum of  $n$ th roots of unity = 0; Product =  $(-1)^{n-1}$
- Geometric view: roots lie equally spaced on a circle, forming a regular polygon
- Rotation:  $z \cdot e^{i\theta}$  rotates  $z$  by  $\theta$  radians anticlockwise about the origin

*These are exactly the tools used to solve every question in Exercise 2.8*

